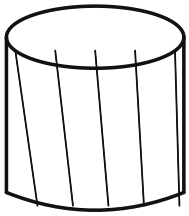
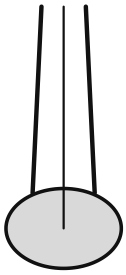
PHYSICAL AND CHEMICAL SCIENCE

# Sound

Vibrations, decibels, how sound travels, sound insulation, and pitch in musical instruments



dhol (drum)



ektara (string)



bansuri (flute)

**In this pack**

📖 **Learn it:** Sound: vibrations, loudness and pitch ★☆☆
**Fact check:** What vibrates? ★☆☆
**Think it through:** Thinking about sound ★★☆
**Investigate:** Which material blocks sound best? ★★☆
**Real-world case study:** Too much noise! ★★★
**Review and challenge:** Show what you know ★★★

**This pack belongs to:** _____ **Class:** _____

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# Sound: vibrations, loudness and pitch

★☆☆ Grade 3

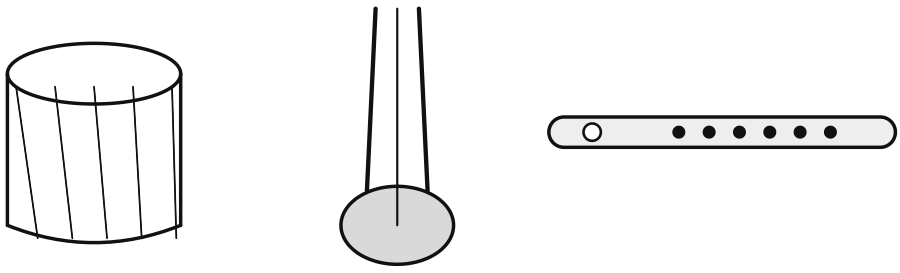
Name: Class: Date:

**Learning goals**

- Explain that sounds are made by vibrations.
- Measure loudness in decibels and describe how sound travels.
- Explain pitch and how musical instruments change it.

Put your fingers on your throat and hum. Can you feel a tingle? That is a **vibration**: a very fast shaking. **Every sound is made by something vibrating.**

- When you hit a **dhol**, its skin vibrates.
- When you pluck an **ektara**, its string vibrates.
- When you blow a **bansuri**, the air inside it vibrates.
- When you talk, your **voice box** in your throat vibrates.

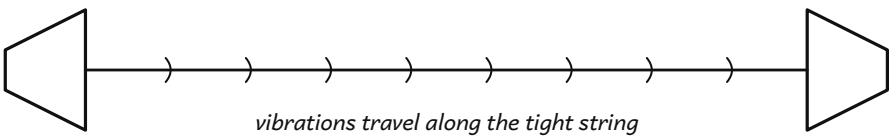


**dhol (drum)** **ektara (string)** **bansuri (flute)**

*Instruments from Bangladesh.*

## How sound travels

The vibrations pass from one tiny particle to the next until they reach your **ear**. Sound can travel through **gases** (air), **liquids** (water) and **solids** (wood, metal, string). It travels better through solids: put your ear on a desk while a friend taps the other end! Sound cannot travel through empty **space**, where there is no air.

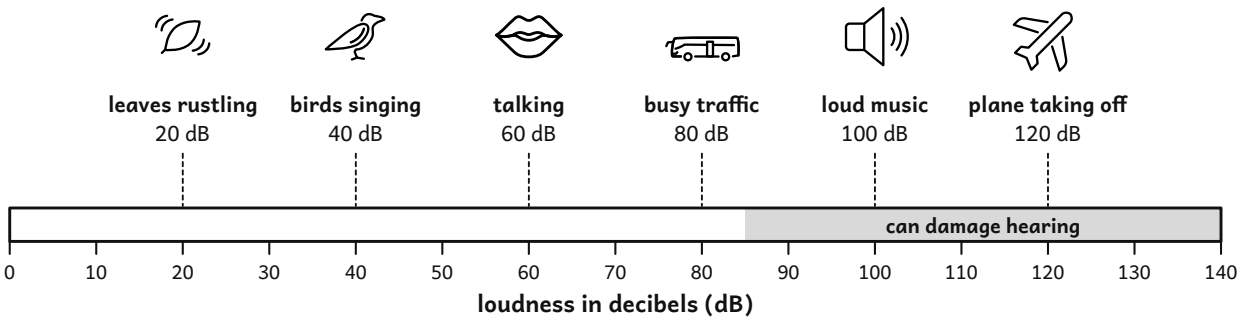



*A string telephone.*

## How loud?

**Loudness** (volume) is measured in **decibels (dB)** with a **sound level meter**. Many mobile phones have a sound meter app. Sounds louder than about **85 dB** can damage

your hearing if you listen for a long time.



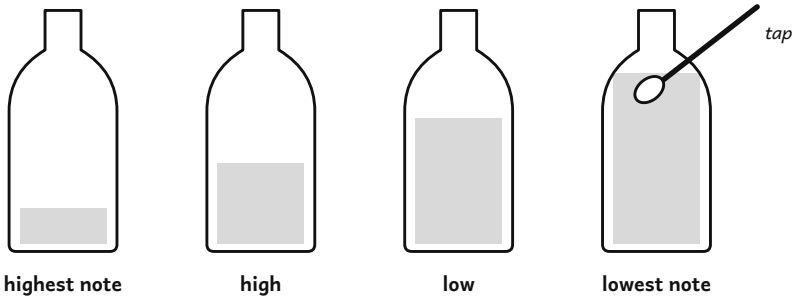

*Everyday sounds on the decibel scale.*

Soft, thick materials such as **foam**, **cloth**, **curtains** and **egg boxes** soak up sound. They stop it from travelling through. This is called **sound insulation**. Ear defenders use it to protect workers' ears.

## High and low: pitch

**Pitch** is how **high** or **low** a sound is. A bird tweets with a high pitch; a big drum booms with a low pitch. High and low sounds can each be loud or soft.

| Higher pitch | Lower pitch |
|---|---|
| short, thin or tight strings | long, thick or loose strings |
| small drums, tight drum skins | big drums, loose drum skins |
| short tube of air (cover fewer holes on a flute) | long tube of air (cover more holes) |




*Tap bottles with different amounts of water.*

**Key words**

| | |
|---|---|
| **vibrate** | shake very fast |
| **volume** | how loud a sound is |
| **decibel (dB)** | the unit for measuring loudness |
| **pitch** | how high or low a sound is |
| **sound insulation** | a material that stops sound travelling through |

FACT CHECK · C2 · SOUND

# What vibrates?

★☆☆ Grade 3

Name: Class: Date:

**Learning goals**

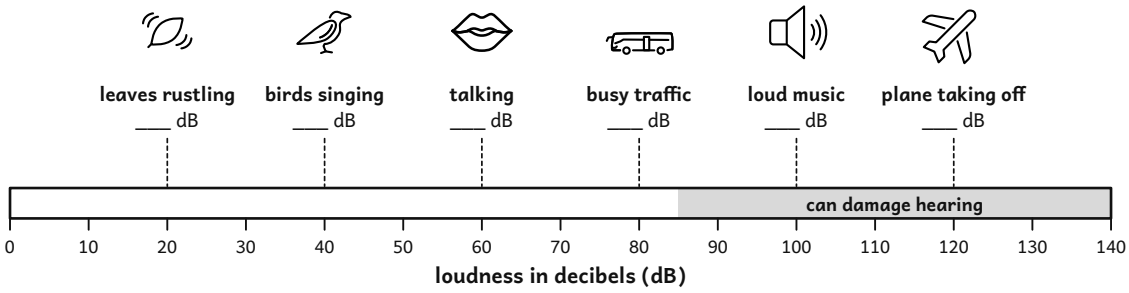
- Name what vibrates.
- Read the decibel scale.

## 1 Draw a line to match the instrument to the part that vibrates. [4]

Draw a line to match each pair, or write the letter.

| | |
|---|---|
| dhol ● | ● **A** the drum skin |
| ektara ● | ● **B** the air inside |
| bansuri ● | ● **C** the string |
| your voice ● | ● **D** the voice box in your throat |

## 2 Write the loudness of each sound. [3]




| leaves | birds | talking | traffic | loud music | plane |
|---|---|---|---|---|---|
| | | | | | |

## 3 True or false? [3]

| Statement | True | False | If false, correct it |
|---|---|---|---|
| 1. Sound can travel through water. | | | |
| 2. Sound travels through space. | | | |
| 3. A loud sound always has a high pitch. | | | |

THINK IT THROUGH · C2 · SOUND

# Thinking about sound

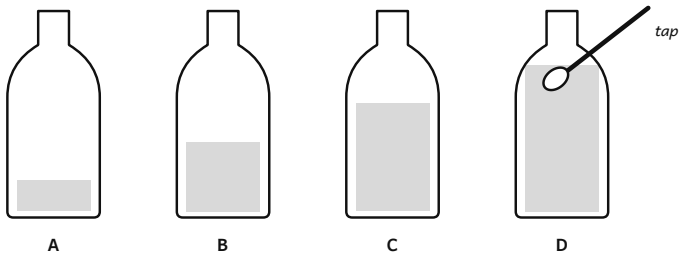
★★☆
Grade 3

Name: Class: Date:

**Learning goals**

- Explain pitch and loudness.

**1** **Which bottle makes the highest note when tapped? Which makes the lowest?** [2]




**2** **An ektara player presses the string to make it shorter. What happens to the pitch?** [1]

**3** **Why do people who work with loud machines wear ear defenders?** [2]

**4** **A string phone only works when the string is tight. Why?** [2]

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# Which material blocks sound best?

★★☆ Grade 3

Name: Class: Date:

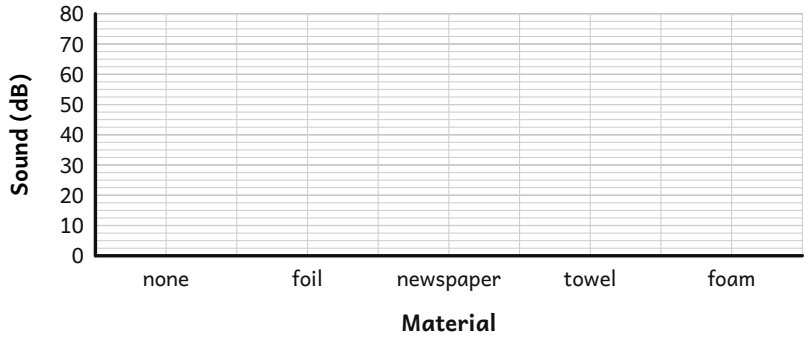
**Learning goals**

- Measure sound in decibels.
- Draw a bar chart and conclude.

Class 3C put a ringing alarm clock inside a box and wrapped it in different materials. They measured the sound 1 metre away with a sound meter app.

**1** **Draw a bar chart.** [3]

| Material | none | foil | newspaper | towel | foam |
|---|---|---|---|---|---|
| Sound (dB) | 70 | 66 | 60 | 50 | 45 |




**2** **Which material blocked the sound best?** [1]

**3** **What kind of materials are good sound insulators?** [2]

**4** **What did they keep the same to make it fair?** [1]

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REAL-WORLD CASE STUDY · C2 · SOUND

# Too much noise!

★★★
Grade 3

**Name:** **Class:** **Date:**

**Learning goals**

- Use what you know about sound to solve a real problem.

Science | Health | Citizenship

Dhaka is one of the noisiest cities in the world. On busy roads the sound is often above **85 dB** because of car horns, buses, building work and loudspeakers. Traffic police stand in this noise all day.

Too much noise can damage hearing and make it hard to sleep, learn and think. Bangladesh has rules: very loud hydraulic horns are not allowed, and roads near **schools and hospitals** are **silent zones** where drivers must not honk.

Based on reports about noise pollution in Dhaka and Bangladesh's noise rules.

**1** **Name three things that make Dhaka noisy.** [3]

**2** **Why are roads near hospitals and schools silent zones?** [2]

**3** **Make a "Do not honk" sign for your school gate.** [2]

Silent zone sign

REVIEW AND CHALLENGE · C2 · SOUND

# Show what you know

★★★
Grade 3

Name: Class: Date:

**Learning goals**

- Use sound words.

## 1 Find the words.

| J | U | L | O | U | D | V | Y | P |
|---|---|---|---|---|---|---|---|---|
| E | S | L | W | V | T | V | G | I |
| H | A | U | Y | I | W | K | D | T |
| D | I | R | U | B | T | G | E | C |
| N | R | L | G | R | I | N | C | H |
| O | S | U | R | A | M | B | I | I |
| I | V | O | M | T | H | W | B | U |
| S | K | G | W | E | B | B | E | K |
| E | N | J | B | V | D | O | L | H |

**decibel drum ear loud noise pitch vibrate**

## 2 Loudness is measured in... [1]

◯ metres ◯ decibels

◯ grams

## 3 A short, tight string makes a... [1]

◯ high pitch ◯ low pitch

◯ no sound

## 4 Challenge. Design a quiet corner for your classroom. Which materials would you use? [2]

.....

.....

**How am I doing? Colour or tick a circle.**

| I can... | Not yet | Nearly | Got it! |
|---|---|---|---|
| explain that sounds come from vibrations | ○ | ○ | ○ |
| measure sound in decibels | ○ | ○ | ○ |
| say which materials block sound | ○ | ○ | ○ |
| explain pitch | ○ | ○ | ○ |
| change the pitch of an instrument | ○ | ○ | ○ |